

Question				Marking details		Marks available					
						AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		as pH increases, citric acid content decreases and sugar content increases	✓			2	2	2	
				as acidity decreases, ascorbic acid content decreases and water content decreases							
				tomatoes are a good source of vitamin C and citric acid							
				citrus fruits contain ascorbic acid and a natural preservative	✓						
		(ii)		award (1) for any of following - the values for ascorbic acid are bigger but the unit is 1000 times smaller - mg/100g is a smaller unit than % - citric acid is measured in % but ascorbic acid is measured in mg/100g - there is much more citric acid (than ascorbic acid) present				1	1	1	
	(b)	(i)		$\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$ formulae (1) state symbols (1)		2			2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)	I	$0.35 \times 0.021 = 0.00735$		1		1	1	1
			II	$0.588 / 0.59$ (2) if answer incorrect credit each correct step in method $0.00735 \times 2 = 0.0147$ (1) $\frac{0.0147}{0.025} = 0.588 / 0.59$ (1) ecf possible from part (i)		2		2	2	2
		(iii)		1.2×10^{22} (2) accept $12 \times 10^{21} / 0.12 \times 10^{23}$ if answer incorrect credit each correct step in method but do not award any marks for multiplying mass 0.36 by N_A $\frac{0.36}{18} = 0.020$ (1) $0.020 \times 6.0 \times 10^{23} = 1.2 \times 10^{22}$ (1)		2		2	2	
				Question 8 total	2	5	3	10	8	3